



Practical Solutions to Protect Data Privacy

Data Saturday Columbus

August 16, 2025

We'll cover:

Defining Data Privacy

- Why it's Important

Disclosure Risk

Privacy-enhancing techniques

- Masking
- Differential Privacy
- Text redaction and replacement
- Synthetic Data

Comparisons

Hi!

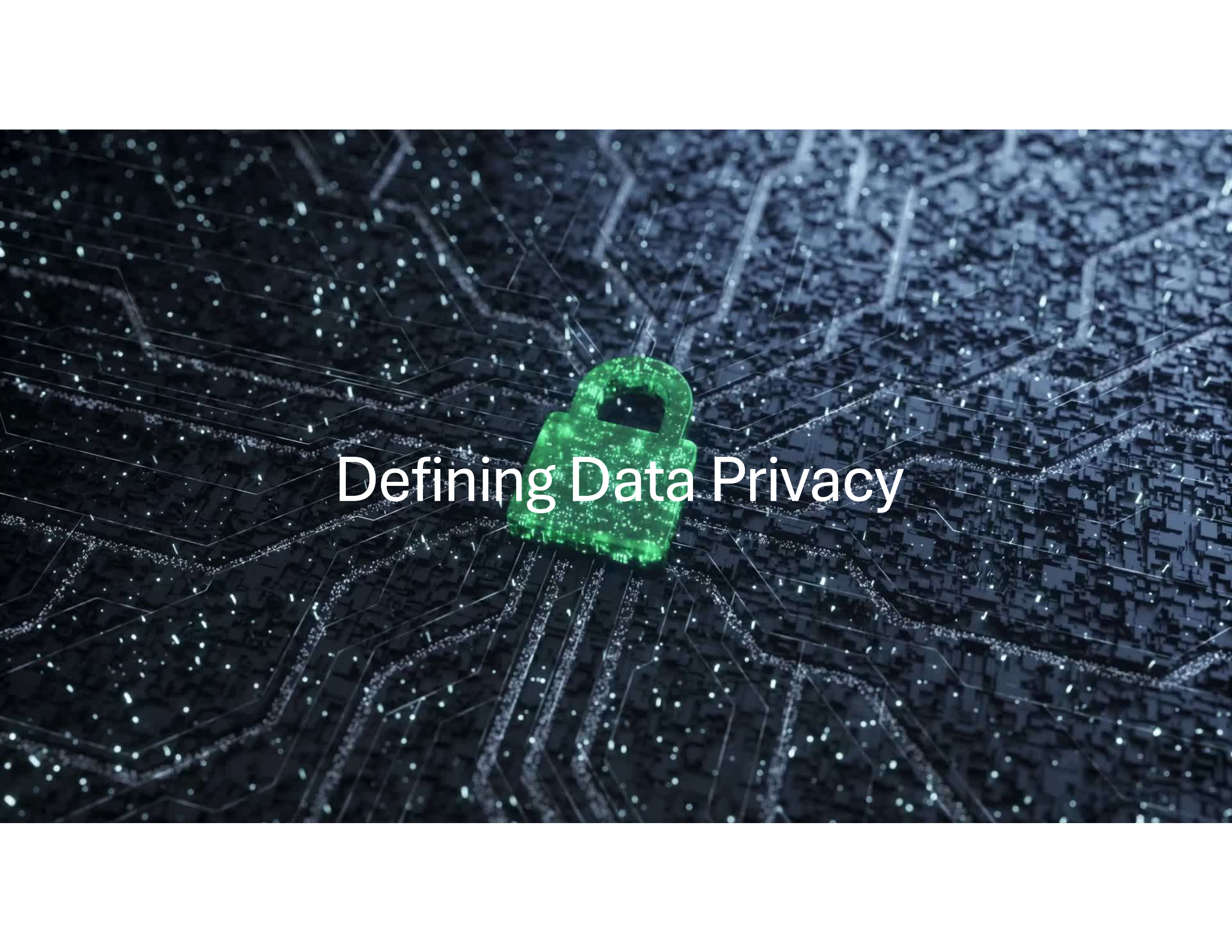
Britton Gray

- Data professional for almost 20 years
 - My passions
 - Advanced analytics (data and language)
 - Data privacy
 - Data engineering

I'm an actual scrum master!



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SOLUTIONS

The background is a dark, high-contrast image of a circuit board. The traces of the board are highlighted in a bright blue or cyan color, creating a complex, web-like pattern. In the center of the image, there is a glowing green padlock. The padlock has a textured, almost crystalline appearance, with light reflecting off its surfaces. It is positioned slightly to the left of the center, and the text 'Defining Data Privacy' is overlaid on it.

Defining Data Privacy

Data Privacy (*n*):

Any information about an **identified or identifiable** natural person

Data privacy rights:



Who has access to their information



What information can be accessed



When: Data kept only as long as it's legitimately needed



Where: Limitations on sharing and where data is stored



Why: the purpose(s) for storing/accessing that information



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Legitimate Purposes

Valid reasons for storing and processing data (GDPR)



Affirmative consent of the data subject



Vital interests of a person



Fulfillment of a contract



Legal obligation



In the public interest



Legitimate interests of processor – balanced by rights of the data subject

Why is this important?

Ethics!

Legal sanctions

- Fines
 - \$2500-\$7500 per person affected under some state laws
 - GDPR: Max €20 million / 4% of annual turnover
- Consent decrees
- **Private lawsuits**

Disclosure Risk

CONFIDENTIAL



Disclosure Types

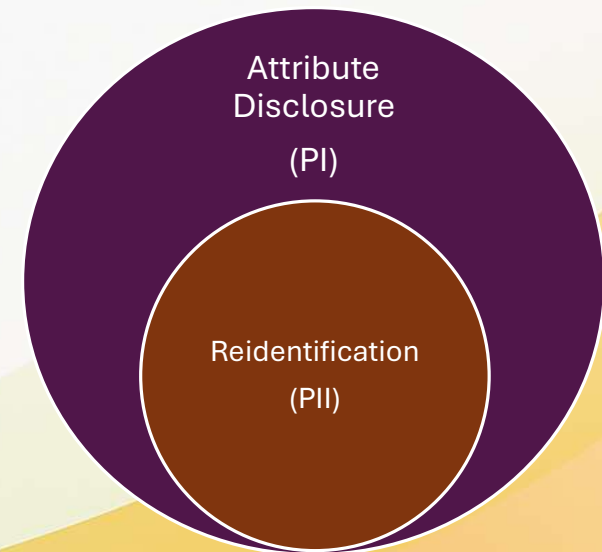
- **Reidentification**
- A specific person is identifiable

Attribute Disclosure

One or more sensitive attributes or behaviors about a person become discoverable

“100% of our students met standards on the state exam!”

- Actual school who sued the state because they couldn't report it



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“Anonymized” Medical Record

All identifiers have been removed

Hospital	162: Sacred Heart Medical Center in Providence
Admit Type	1: Emergency
Type of Stay	1: Inpatient
Length of Stay	6 days
Discharge Date	Oct-2011
Discharge Status	6: Dsch/Trfn to home under the care of a health service organization
Charges	\$71,708.47
Payers	1: Medicare 6: Commercial insurance 625: Other government-sponsored payers
Emergency Codes	E8162: motor vehicle traffic accident due to loss of control; loss control mv-mocycl

Diagnosis Codes	80843: closed fracture of other specified part of pelvis 51851: pulmonary insufficiency following trauma & surgery 86500: injury to spleen without mention of open wound into cavity 80705: closed fracture of rib(s); fracture five ribs-close 5849: acute renal failure; unspecified 8052: closed fracture of dorsal [thoracic] vertebra without mention of spinal cord injury 2761: hyposmolality &/or hyponatremia 78057: tachycardia 2851: acute posthemorrhagic anemia
------------------------	--

Age in Years	60
Age in Months	725
Gender	Male
ZIP	98851
State Reside	WA
Race/Ethnicity	White, Non-Hispanic
Procedure Codes	5781: Suture bladder laceration 7939: 7919: Open/Closed reduction of fracture of other specified bone
Physicians	...
...	...



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Newspaper Article

MAN, 60, THROWN FROM MOTORCYCLE

A 60-year-old Soap Lake man was hospitalized Saturday afternoon after he was thrown from his motorcycle. Ronald Jameson was riding his 2003 Harley-Davidson north on Highway 25, when he failed to negotiate a curve to the left. His motorcycle became airborne before landing in a wooded area. Jameson was thrown from the bike; he was wearing a helmet during the 12:24 p.m. incident. He was taken to Sacred Heart Hospital. The police cited speed as the cause of the crash.

(name changed)

[News Review 10/18/2011]



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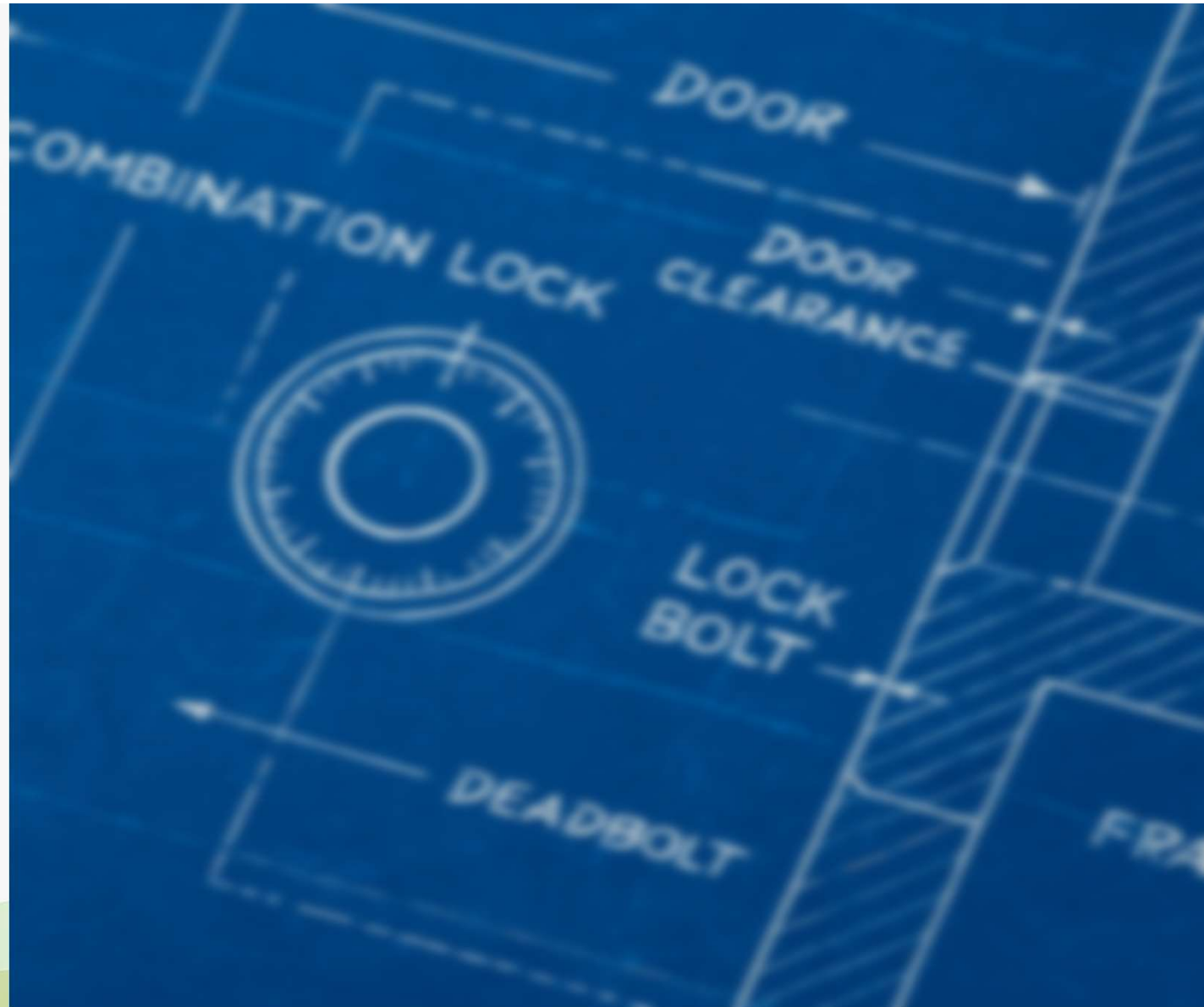
Reidentification → Attribute Disclosure

Record	000000000
Hospital	162: Sacred Heart Medical Center in Providence
Admit Type	1: Emergency
Type of Stay	1: Outpatient
Length of Stay	6 days
Discharge Date	Oct-2011
Discharge Status	under the care of an health service organization
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Age in Months	720
Gender	Male
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State Reside	WA
Race/Ethnicity	white, Non-Hispanic

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Credit: "Only You, Your Doctor, and Many Others May Know"
Journal of Technology Science
Dr. Latanya Sweeney, 2015

Techniques



Masking

A photograph of a tablet screen showing a table with two columns: 'Email' and 'State'. The email addresses are masked with asterisks. The states listed are Indiana, California, Texas, New York, Florida, Michigan, Illinois, and Pennsylvania.

Email	State
b***@g***.com	Indiana
j****@y***.com	California
m***@o***.com	Texas
s***@g****.com	New York
p***@y***.com	Florida
a***@o***.com	Michigan
k***@y***.com	Illinois
t***@g***.com	Pennsylvania

Masking

- Useful for individual records
 - Be aware of some aggregation issues (GROUP BY)
- Works with column / user / row-level security
- Two types of masking targets:
 - Would-be or could-be identifiers
 - Sensitive attributes
- Efficacy heavily depends on what and how you mask

Masking Methods

Mask	Example	Usage Notes
Redaction	Britton → NULL	No need to do any analysis on the field, NULL may convey mistaken meaning.
Constant Masking	Britton → ****	Same as above, but mask is obvious
Partial Masking	bgray@pltw.org → b***@p***.org 8/6/2024 → 1/1/2024 12.34.56.78 → 12.34.56.xx	Decreases reidentification risk, but doesn't eliminate it Useful when some aggregation needed
Hashing	Britton → 1e00fea58e20cc1	Works with aggregates. Sliceable data that's obviously masked. Salt the hash: ("AA1438-Britton")
Banding / Blurring	33 → 25-34 (discrete) 33 → 30 (aggregable)	Numeric variables important to analysis but pose reidentification risk
Geo Blurring	39.9155°N, 86.0650°W 39.92°N, 86.07°W	Treat precise geo data as identifying. 1/100 of a degree = about ½ mile

Masking in Data Platforms



Implemented via **masking policies**

- Define once, use multiple times
- Anything you can do in a CASE statement



Configurable pre-defined masking functions, applied to columns

- Implemented in Fabric's SQL analytics endpoint and Warehouse



Not native; ideally use OLS / RLS

Idea: Source view, UNION masked and unmasked, use RLS DAX



Same as Power BI

- Understand who can see what

Differential Privacy

Adding noise



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A quick game

Find the salary:

```
SELECT
  AVG(SALARY) ,
  COUNT(*)
FROM EMPLOYEES
WHERE . . .
```

Do so with....

2 Queries, return 5+ rows

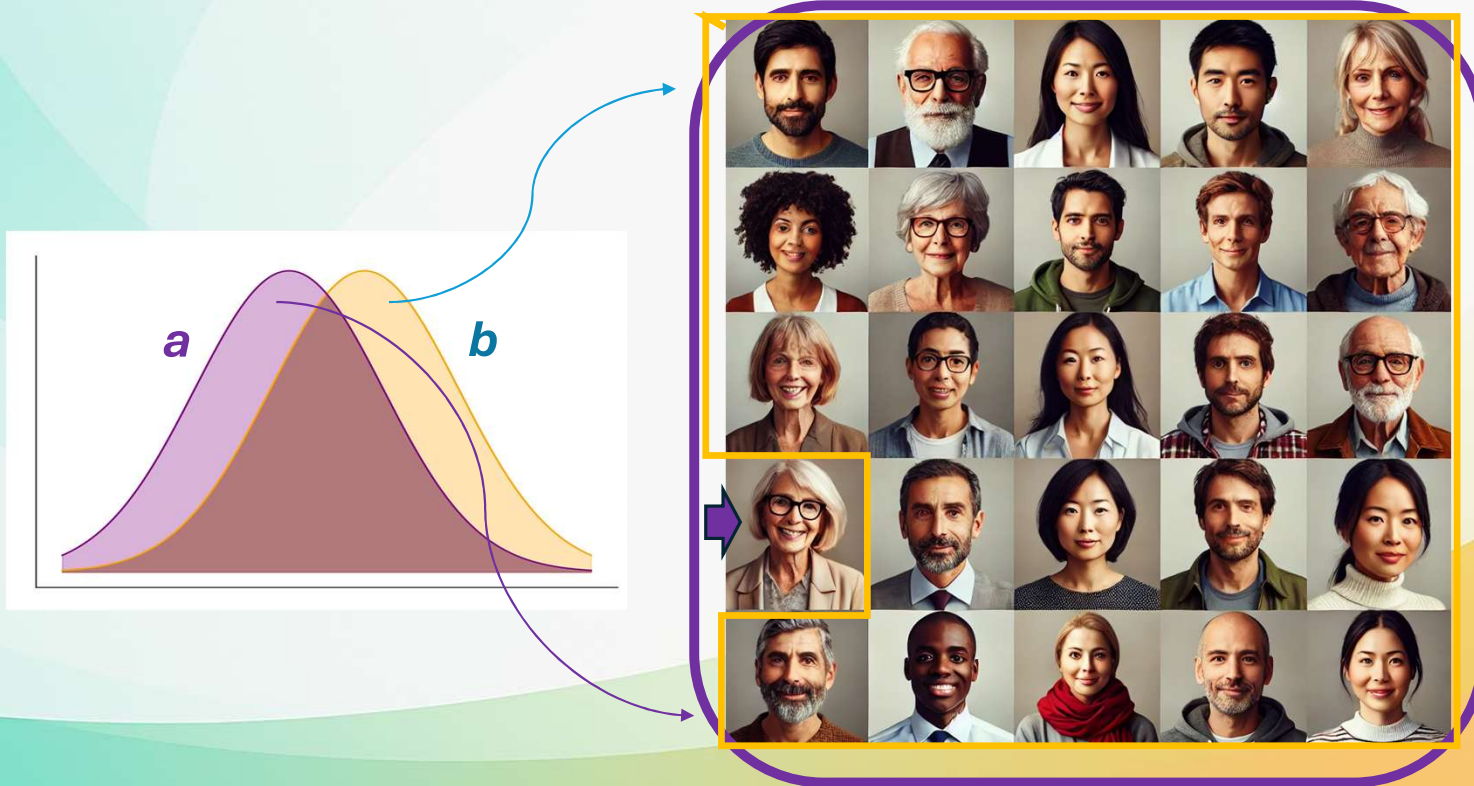




(specifically, a **differencing attack**)

Differential Privacy: Definition

- Adding statistical perturbation (noise) to aggregated query results



a
is indistinguishable
from *b*

The What & Why Differential Privacy

- Used in aggregate queries only
- Guards against differencing attacks
- It's still possible to uncover protected attributes through brute force
 - Countermeasures: **privacy budgets and results caching**

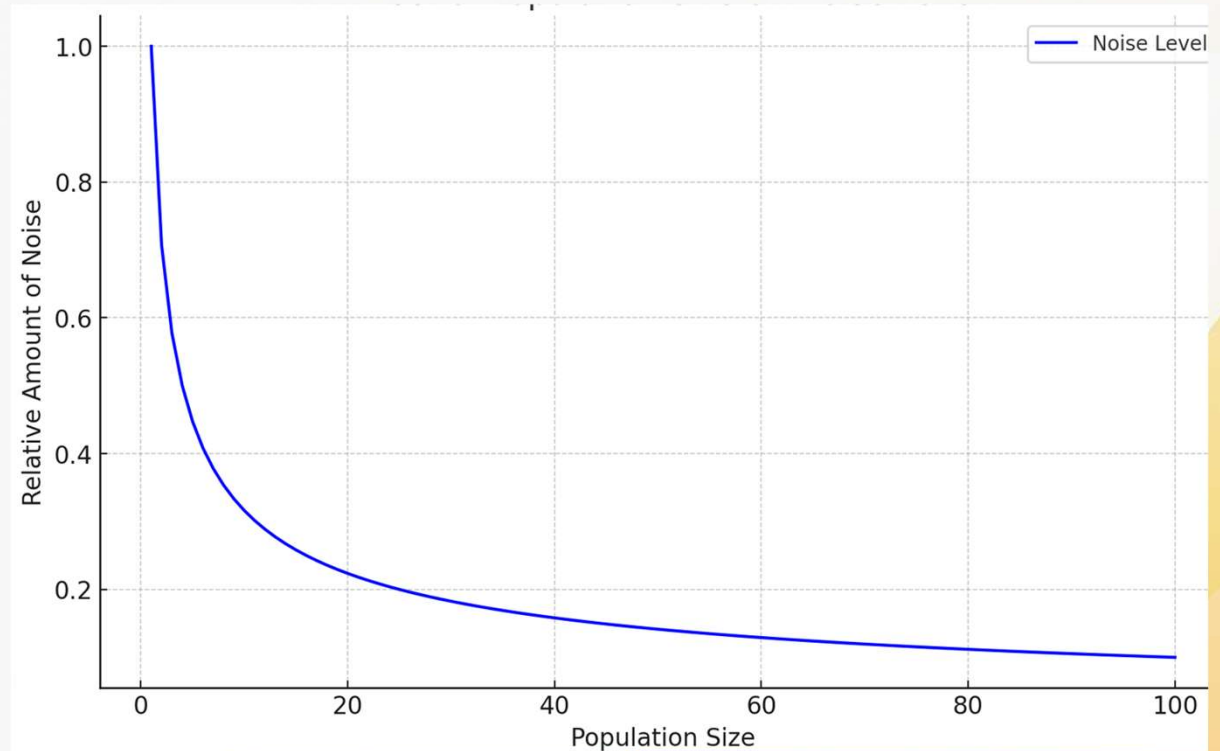


A person is seen from the side, looking at a computer monitor. The monitor displays a table with two columns: 'Age Range' and 'Population'. The table contains two rows of data. The first row shows '30-38' for the age range and '129 ± 8' for the population. The second row shows '30-39' for the age range and '127 ± 8' for the population.

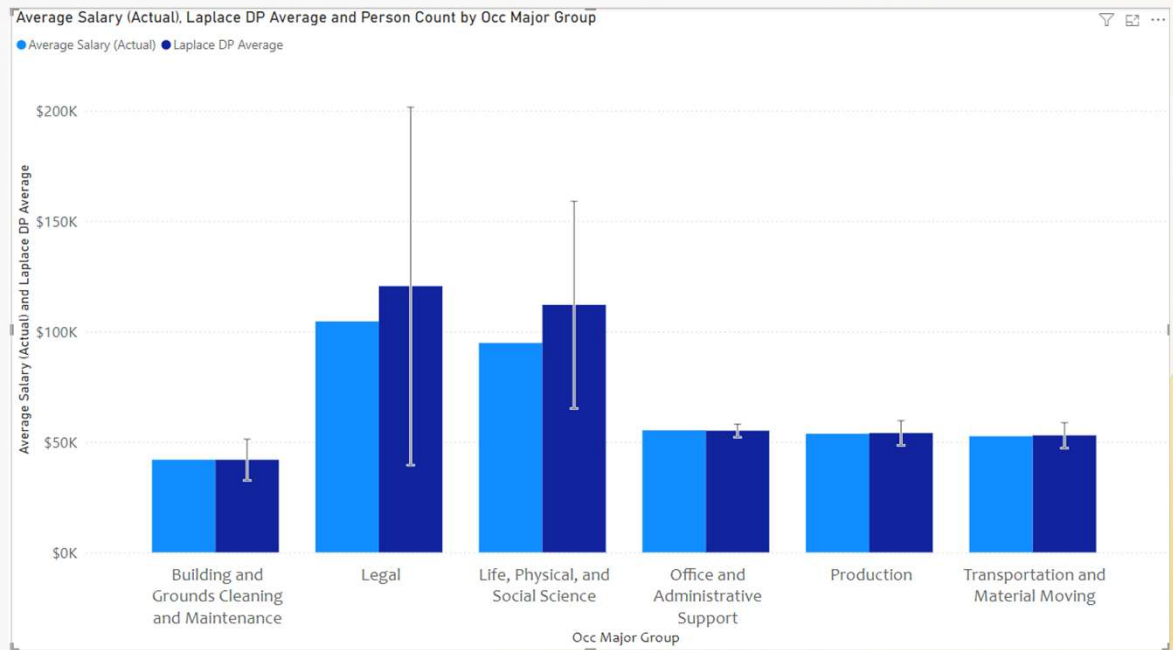
Age Range	Population
30-38	129 ± 8
30-39	127 ± 8

When Using Differential Privacy

- Phrase results as an interval
- Average salary: \$90,538 (\$88,517 - \$92,559)
- More people = narrower range
- True differential privacy will infuse some “empty cells” with data



Diff Privacy: Example



Job Category	# People	Estimate Width
Legal	8	\$162K
Maintenance	51	\$19K
Office & Admin Support	240	\$6K

Differential Privacy: Data Platforms



Snowflake: implemented as a privacy policy

Designed with active privacy attacks in mind, so it's rigid

Currently has several limitations; makes machine SQL difficult



May be third party solutions

- Can use in-database Python with PyDP library



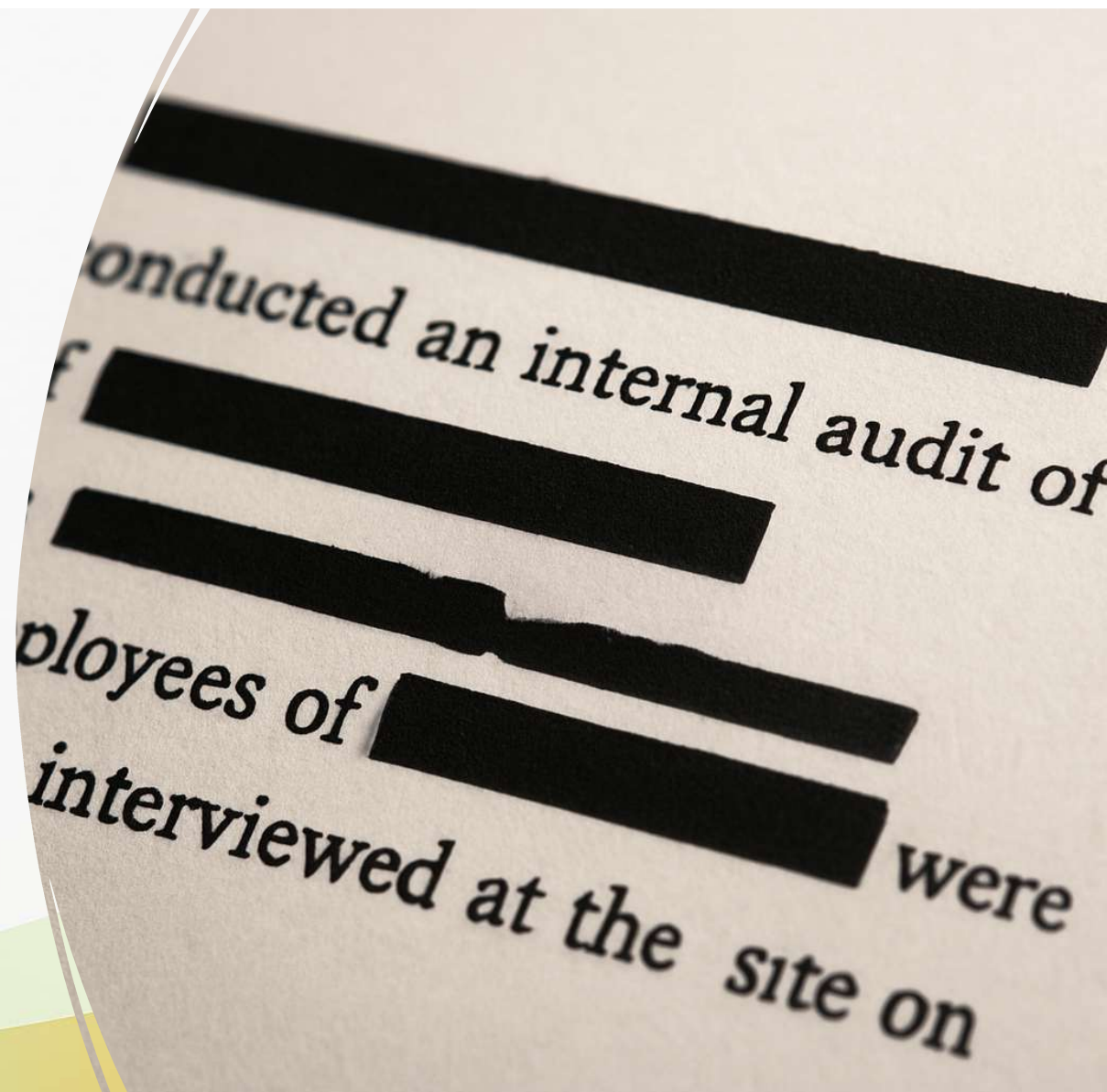
Can be approximated in both technologies

- More useful against accidental disclosure
- Will not be safe from an active privacy attack



Redaction

Preserving privacy in
qualitative data



Qualitative Data



Our native language: stories, conversations, context



Explains the *why* behind numbers and trends



Found in surveys, transcripts, documents, and messages



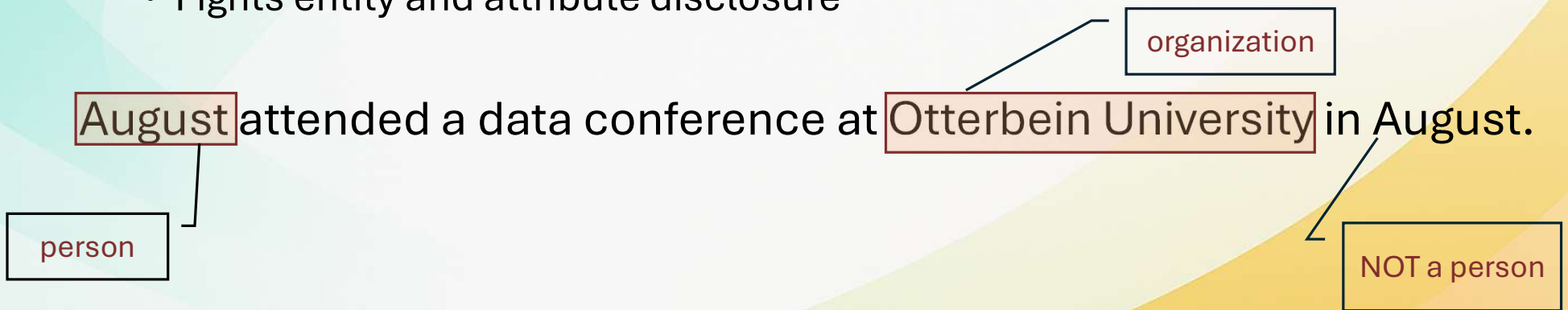
Enables questions like: “*What did our detractors say?*”



AI unlocks analysis at scale

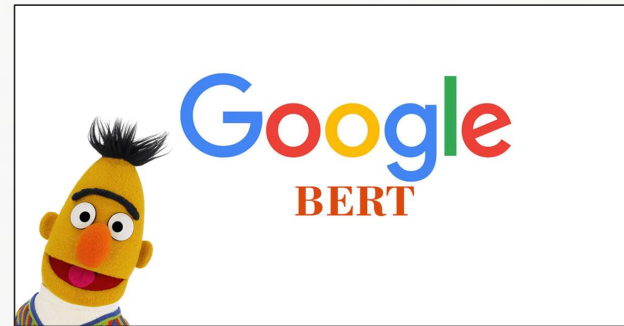
Automated Redaction

- Uses language models to find PII
 - **Named Entity Recognition (NER)**
 - Small Language Models that look at words **in context**
 - Fights entity and attribute disclosure



Common Tools

- Implemented via **Small Language Models** (SLMs)
- BERT is a family tree
 - DistilBERT-NER
 - RoBERTa
 - and many more
- All support fine-tuning
- Generally, > 90% accuracy



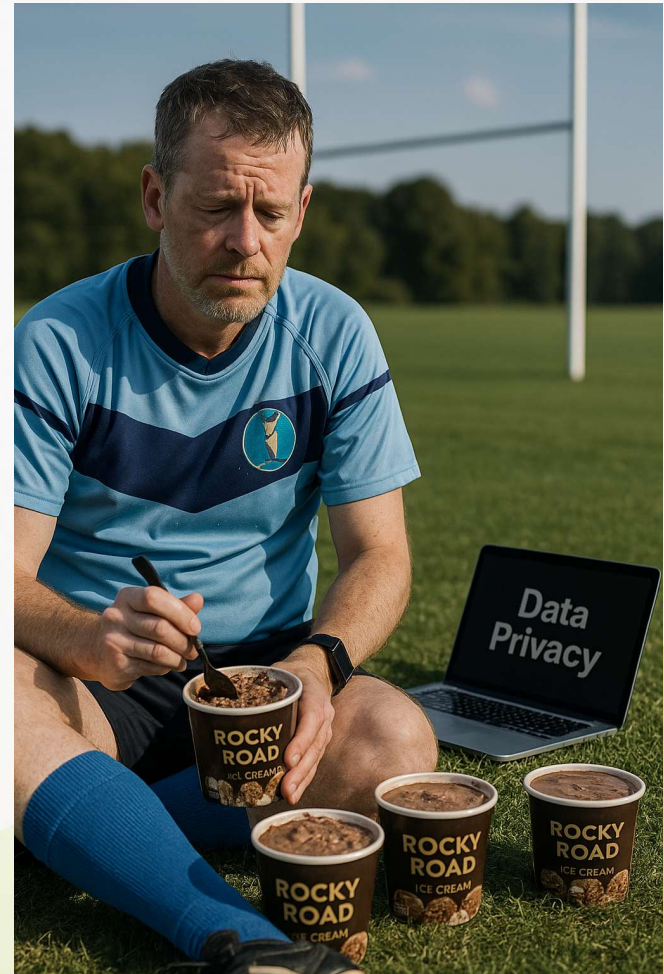
spaCy

Redaction Recommendations

- Integrate redaction into data pipeline
- Redact with a fixed identifier or number of characters
- If you DIY, chain models for maximum efficacy
- Grant access only to redacted versions by default

Implicit Identifiers

“A data privacy speaker ate four tubs of Rocky Road ice cream last week after refereeing a rugby match.”



Replacing Implicit Identifiers

Original Sentence:

“A data privacy speaker ate four tubs of Rocky Road ice cream last week after refereeing a rugby match.”

Generalize:

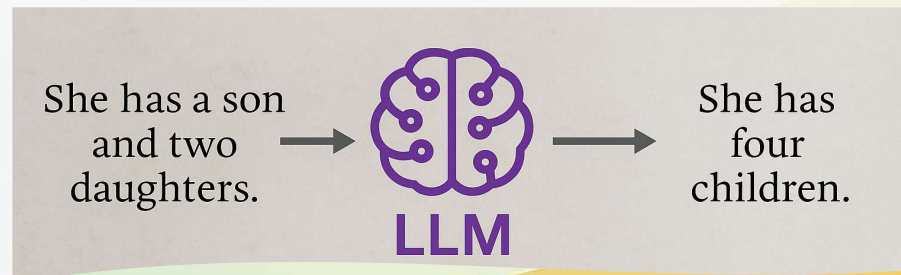
“A technology expert indulged in several desserts last week after participating in a sporting event.”

Modify:

“A cybersecurity analyst ate five cartons of frozen yogurt last month after officiating a soccer game.”

Using LLMs to Replace Implicit Identifiers

- **Can protect against both entity and attribute disclosure**
- LLMs can process text to identify and modify identifiers
- Models constantly improving, but can be expensive
 - Some models can be run locally, e.g., Mistral 7B and GPT-OSS (20B)
 - Running locally may be a plus, as data never leaves your environment



LLM Identifier Replacement: Tips

- Iterate through prompts to find the best instructions
- Add the process early in your data pipeline
- Validate results regularly
- Aggressiveness depends on the model and prompt
- Make sure end users know that text isn't original

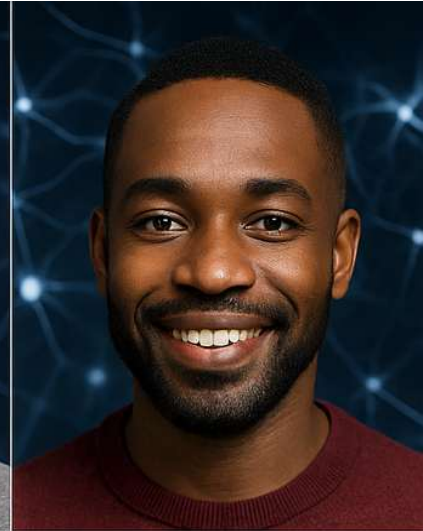
Synthetic Data



Anna Joinsen, 29



Bobly Pirdeser, 41



Blaker McDaug, 35



Bleta Dunner, 63



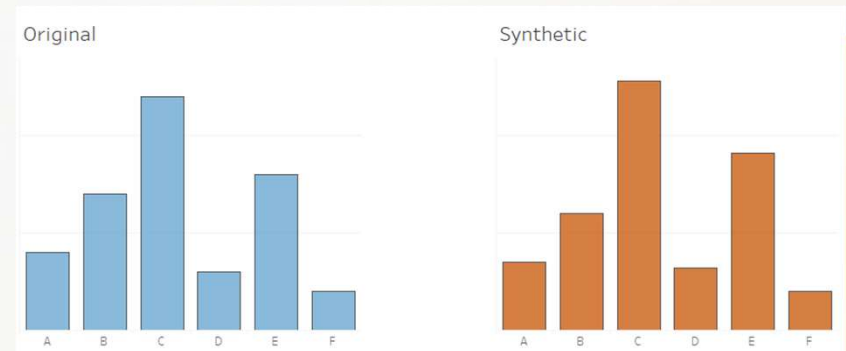
Bee Reiscinbr, 32



Brianton Hard, 46

What is Synthetic Data?

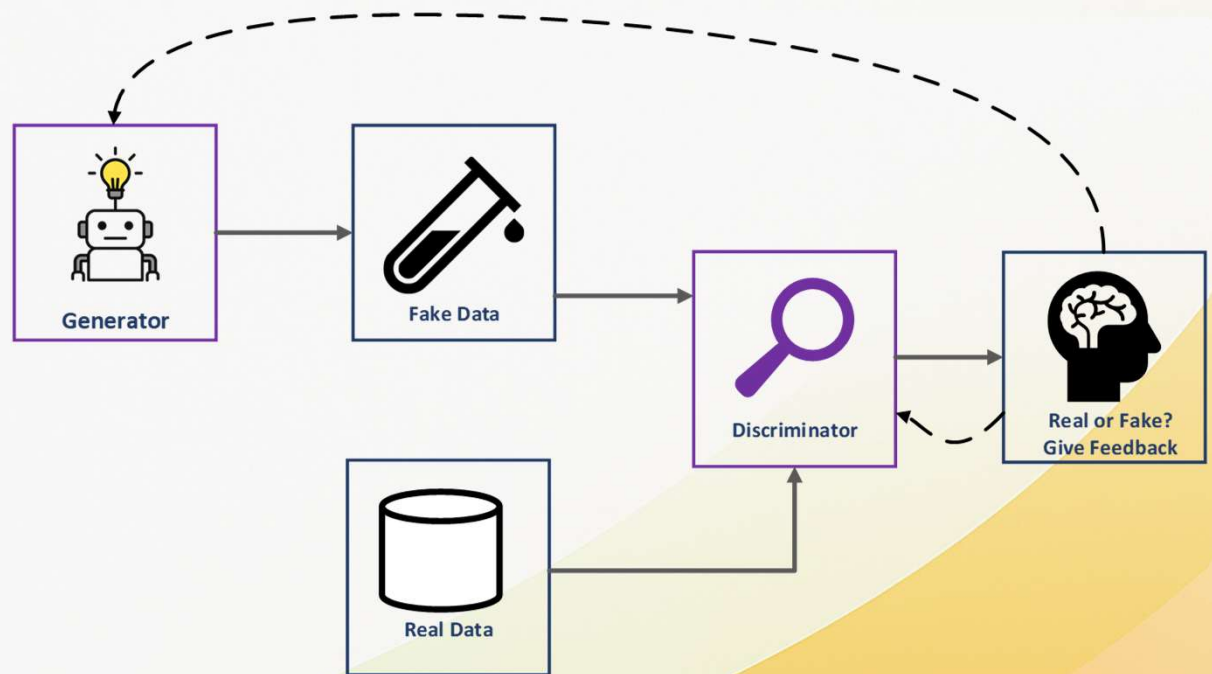
- A new data set generated using real data
- Somewhat statistically consistent
- **Retains no actual source data**
- It can miss rare patterns in data
- No entity or attribute disclosure
- Analyze synthetic data exactly as you would use original data



How is it created?

Most modern tools use **Generative Adversarial Networks (GANs)**

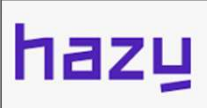
The generator iteratively improves fake data to fool the discriminator trained on real data.



Tools and Platforms

- **Snowflake:** GENERATE_SYNTHETIC_DATA

- **Paid tools** *(some offer free limited-user plans)*



MOSTLY·AI

- **Open-source tools**



- Some require flat data sets, some allow tabular sets

Examples

MOSTLY.AI

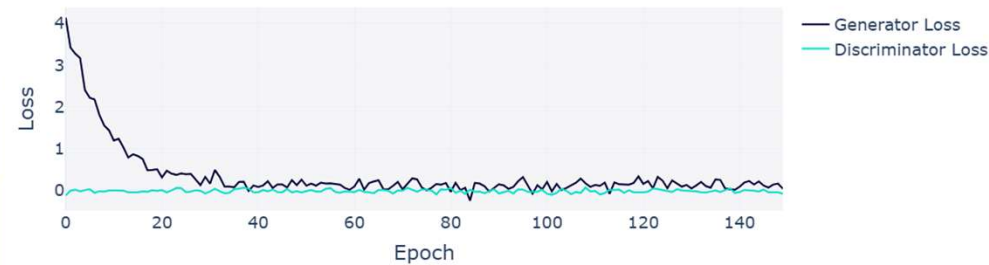
Table columns

Include ⓘ	Name	Encoding type ⓘ
☐	dim_person_key	Primary key
☒	first_name	Language/Text
☒	last_name	Language/Text
☒	type	Tabular/Categorical
☒	gender	Tabular/Categorical
☒	approx_age	Tabular/Numeric: Auto
☒	pop_score	Tabular/Numeric: Auto
☒	address	Language/Text
☒	city	Language/Categorical
☒	state	Tabular/Categorical
☒	zip	Language/Text
☐	latitude	Tabular/Numeric: Auto
☐	longitude	Tabular/Numeric: Auto
☒	dim_occupation_key	Tabular/Categorical
☒	final_salary	Tabular/Numeric: Auto

```
"tables": {
  "table": {
    "columns": {
      "DIM_OCCUPATION_KEY": {
        "sdtype": "categorical"
      },
      "FINAL_SALARY": {
        "sdtype": "numerical",
        "computer_representation": "Int64"
      },
      "ZIP": {
        "sdtype": "postcode",
        "pii": false
      },
      "GENDER": {
        "sdtype": "categorical"
      },
      "APPROX_AGE": {
        "sdtype": "numerical"
      },
      "CITY": {
        "sdtype": "city",
        "pii": false
      }
    }
  }
}
```

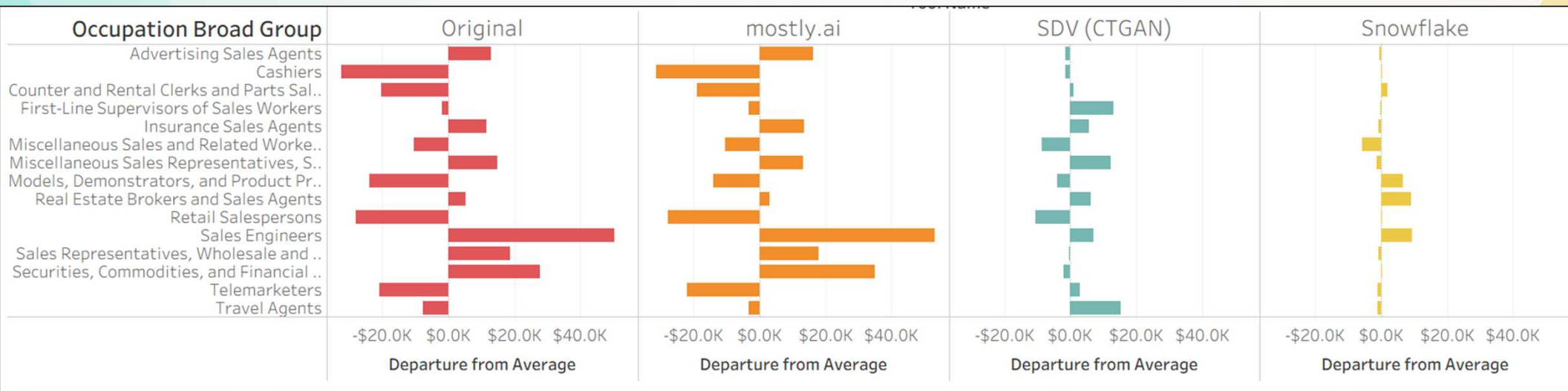


CTGAN loss function



Average Salary Comparisons

By Occupation



Average “Salary” Comparisons

By Gender



Key Insights



Summary of Methods

Tehnique	Best For	Limitation
Masking	Structured, low-complexity data	Can be reversible
Differential Privacy	Aggregate queries	Defeatable via brute force Needs privacy budgets
Redaction (NER)	Transcripts, comments, and other unstructured text that may contain PII	Not 100% accurate Resource-intensive
Replacement	Unstructured text that may contain implicit identifiers	LLMs can get aggressive Very resource-intensive
Synthetic Data	Enabling model training and analytics without using real data.	Outliers not well-represented Some tools better than others

Next Steps

- Assess your data for privacy concerns
- Choose methods based on data type and use case
- Pilot modern approaches to unlock more value
- Build privacy into your data lifecycle: “Privacy by Design”

Thank you!

Questions?

GitHub Repo: https://github.com/IDreamInSQL/data_privacy_resources



 brittongray